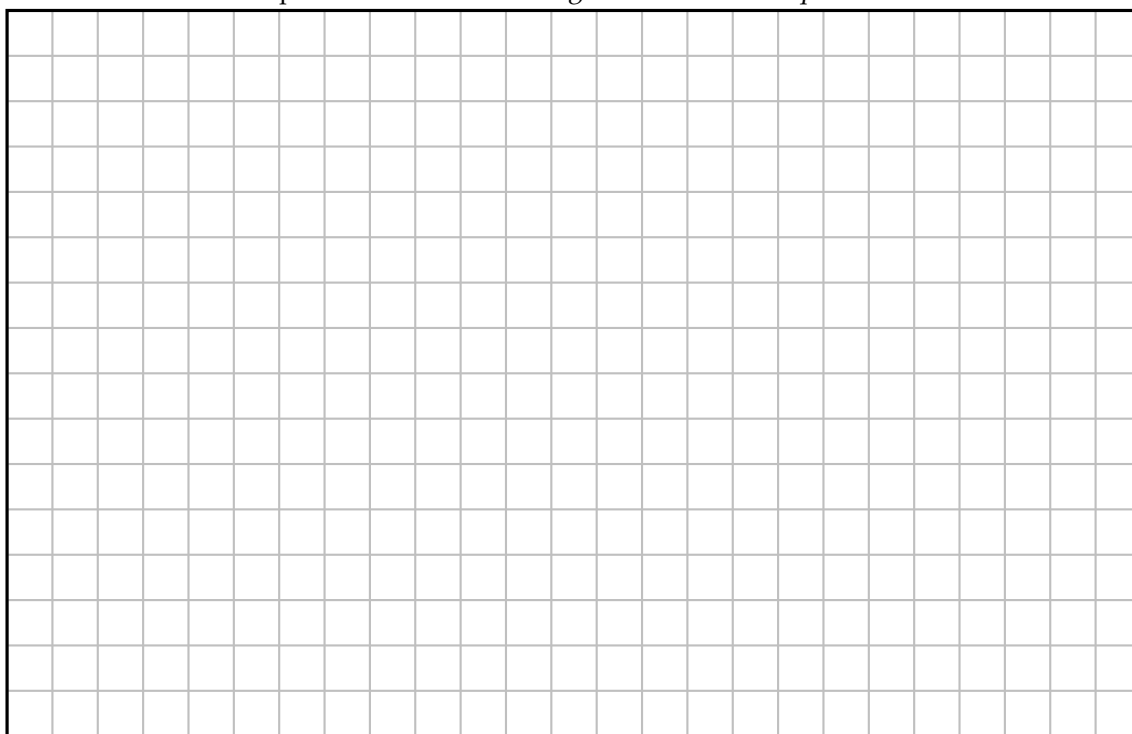


Name: _____ Date: _____

Collaborators: _____ Lab Section: _____

POSTLAB EXERCISES (20 points)*Submit the postlab to the TA at the end of the lab.***Double-Slit Interference and Single-Slit Diffraction (8 points)****Question 3****2 points**

Sketch the observed fringes for double slit interference and single slit diffraction in Graph 1.

Graph 1: *Double-slit and single-slit interference patterns.***Question 4****1 point**Record the double slit separation d , the screen-slit distance L , and the average separation between adjacent fringes y . Be sure to use the same units for all values.Data table 1: *Double-slit measurements.*

Quantity	Measurement	Uncertainty
Double slit separation	$d =$ _____	$\Delta d =$ _____
Screen-slit distance	$L =$ _____	$\Delta L =$ _____
Average fringe separation	$y =$ _____	$\Delta y =$ _____

Question 5**1 point**

Calculate the wavelength λ of the He-Ne laser using the observed double slit pattern and eq. (4).

Question 6**1 point**

Determine the uncertainty $\Delta\lambda$ associated with your experimentally determined wavelength using your estimated uncertainty on L , d , and y . Show your final answer as $\lambda \pm \Delta\lambda$, and compare the result to $\lambda_{\text{He-Ne}} = 632.8 \text{ nm}$. Use the expression

$$\left(\frac{\Delta\lambda}{\lambda}\right) = \sqrt{\left(\frac{\Delta y}{y}\right)^2 + \left(\frac{\Delta L}{L}\right)^2 + \left(\frac{\Delta d}{d}\right)^2} \quad (9)$$

Question 7**1 point**

Which parameter of L , d , and y has the greatest influence on $\Delta\lambda$? Explain your answer.

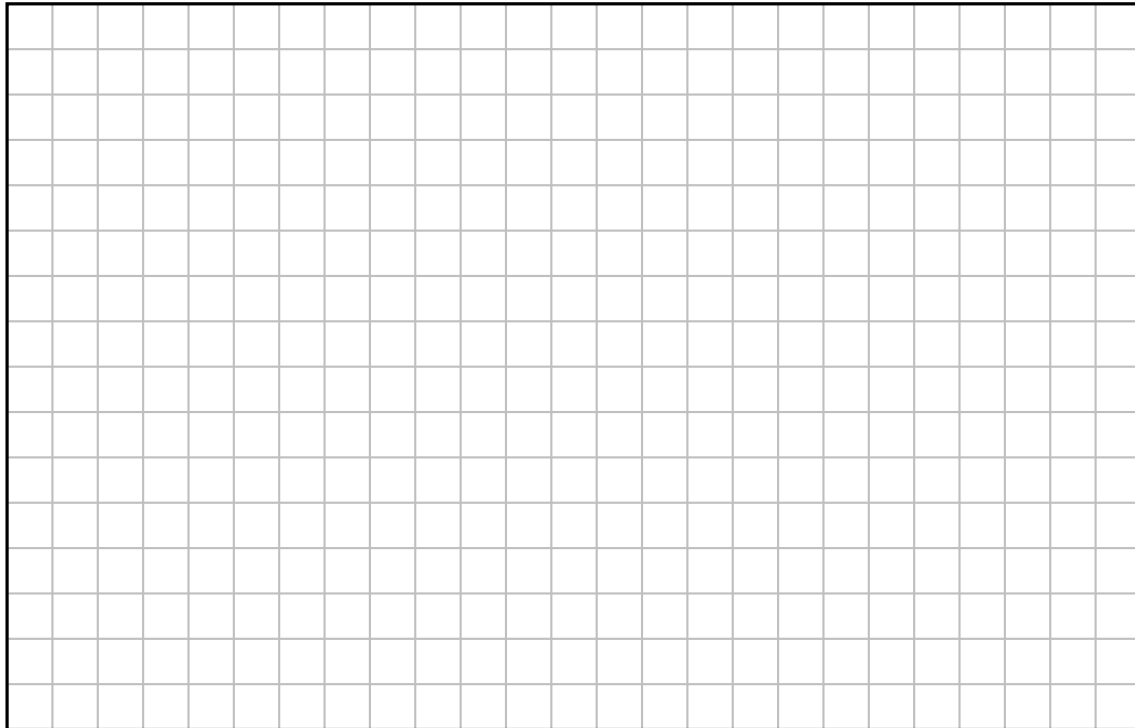
Question 8**2 points**

Compare the single slit and the double slit fringe patterns. What similarities and differences do you observe between them?

Multiple Slit Interference (4 points)**Question 9****2 points**

Sketch the 4-slit and 6-slit interference patterns in Graph 2.

Graph 2: *Multiple slit interference patterns.*

**Question 10****1 point**

What happened to the interference pattern as the number of slits increased? Address issues such as brightness, separation, and sharpness.

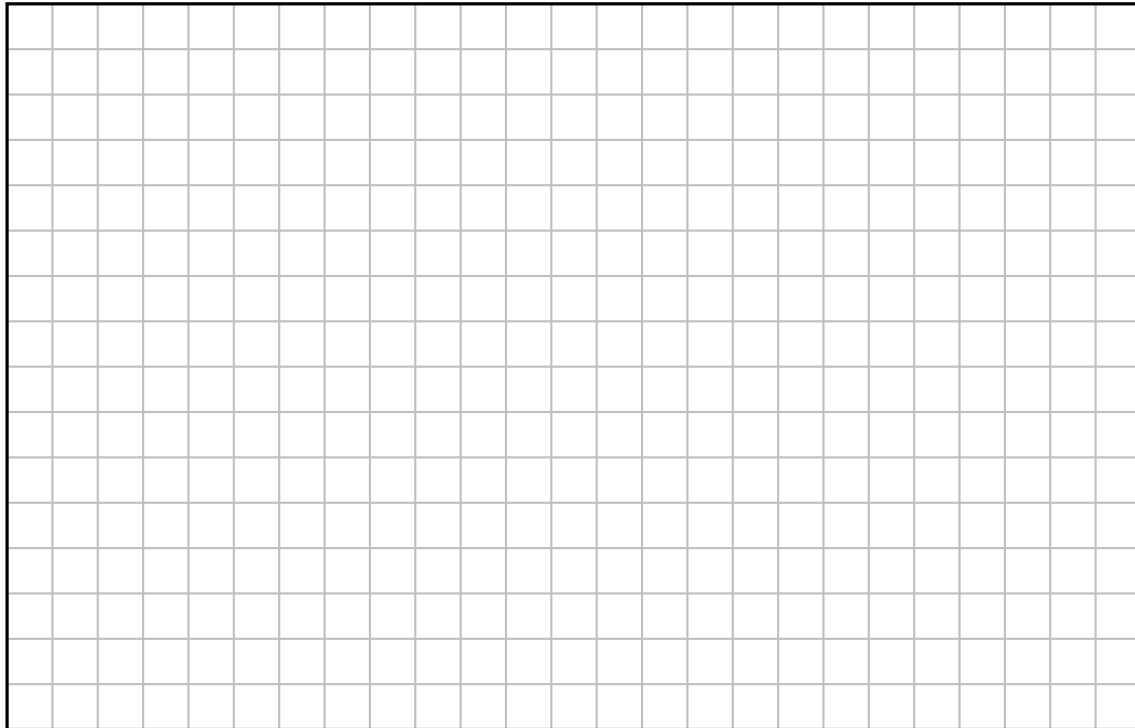
Question 11**1 point**

Without measuring with a microscope, do all the sets of slits on the slide have equal separation? How can you tell from theory and your observations?

Interference by a Wire Mesh (2 points)**Question 12****1 point**

Sketch the interference patterns for the wire mesh in Graph 3.

Graph 3: *Interference pattern for the wire mesh.*

**Question 13****1 point**

How can you create an interference pattern similar to the wire screen interference pattern? Explain your reasoning. If necessary, demonstrate or draw a picture showing how to recreate the wire screen interference pattern.

Microwave Double Slit Measurement (6 points)**Question 14****1 point**

Record the angle and receiver meter reading at 5° increments for as many points as you can obtain. Also record the angles and readings at any maxima and minima you observe.

Data table 2: *Microwave Double Slit Measurements*

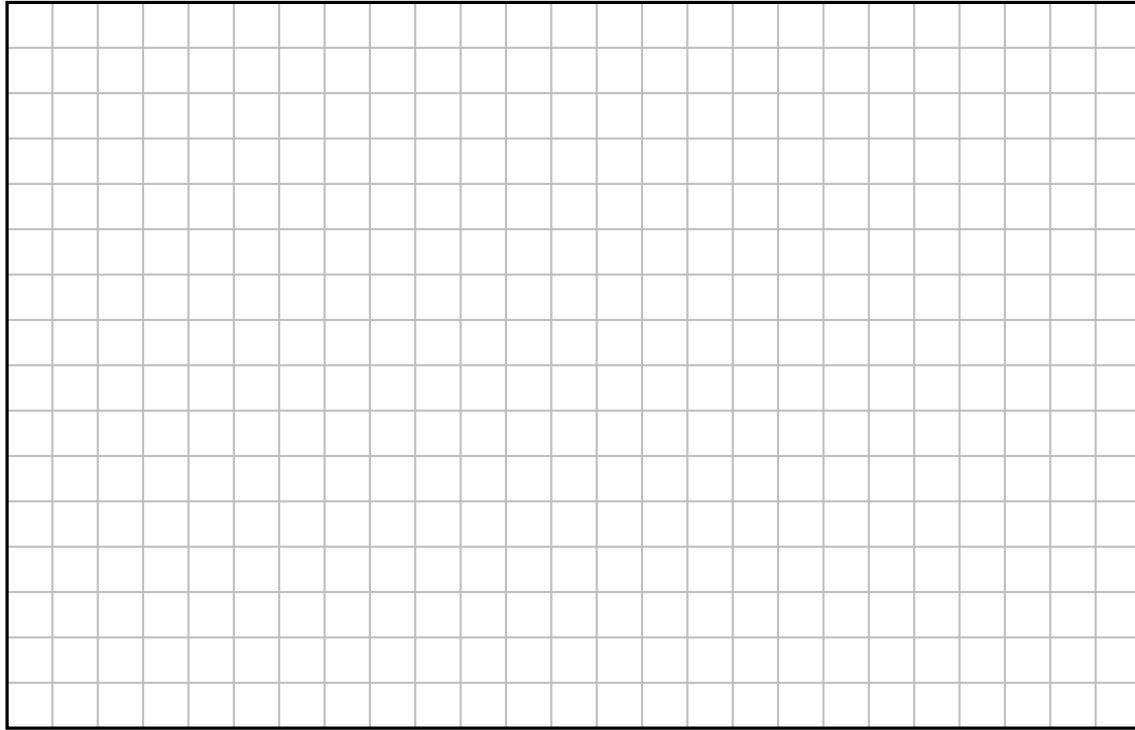
Double slit separation: $d =$ _____ cm

$\theta [^\circ]$	Amplitude	$\theta [^\circ]$	Amplitude	Minima (angle in $^\circ$)	Reading (ampl.)	Maxima (angle in $^\circ$)	Reading (ampl.)
5		-5					
10		-10					
15		-15					
20		-20					
25		-25					
30		-30					
35		-35					
40		-40					
45		-45					
50		-50					
55		-55					
60		-60					
65		-65					
70		-70					
75		-75					
80		-80					
85		-85					
90		-90					

Question 15**2 points**

Plot the amplitude vs. θ on the graph in Graph 4. Include units and labels.

Graph 4: *Microwave double-slit interference pattern.*

**Question 16****1 point**

Using your plot in Graph 4, find the angle between adjacent maxima, θ . You may wish to compute an average value for θ using the distance between several maxima.

Question 17**1 point**

Using your measurement of d , your estimated distance between adjacent maxima θ , and eq. (1), calculate the wavelength λ of the microwave radiation. Record your uncertainties in Data Table 3.

Data table 3: *Double-slit microwave measurements.*

Double slit separation	$d =$ _____	$\Delta d =$ _____
Angle between maxima	$\theta =$ _____	$\Delta \theta =$ _____
Calculated wavelength	$\lambda =$ _____	

Question 18**1 point**

Determine the uncertainty $\Delta\lambda$ associated with your experimentally determined μ -wave wavelength using your estimated uncertainty on d and θ . Show your final answer as $\lambda \pm \Delta\lambda$. Use the expression

$$\left(\frac{\Delta\lambda}{\lambda}\right) = \sqrt{\left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta\theta}{\tan\theta}\right)^2}, \quad (10)$$

expressing the uncertainty $\Delta\theta$ in radians. Note that microwaves are typically around a few centimeters in wavelength.